Semiconductor types

Understanding materials and uses | INVESTMENT / ECONOMY / BUSINESS

Classification by material

Si: the mainstream material for CPUs, GPUs and memory. SiC: high-voltage power control. GaN: power supplies and high-frequency applications. GaAs: used in high-frequency communications and other applications.

Classification by use

Logic: computation. Memory: storage. Power: power control. Analog: signal processing. Sensors: detection of light, temperature, pressure and more.

Avoid confusion

SiC is a material; power semiconductor is a classification by use. Consider materials and uses together.

Use in company analysis

NVIDIA focuses on design; TSMC on contract manufacturing. Evaluate materials, uses and business models separately.

TODOROKI | Investment and economics, guided by numbers rather than intuition.